



Pixelated Transformations: Tracing Changing Irrigation Management Through Deep-Learning-Based Object Detection on HEXAGON and SkySat Satellite Images

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Abstract

Prior applications of deep-learning object detection in archaeology are often critiqued for their limited capacity to generate anthropologically meaningful insights into human societies. We address this by integrating comparative anthropology and fieldwork with a YOLO-based detector to trace shifting irrigation management in Turpan, Xinjiang, at a scale unattainable through manual inspection. Trained on tiled KH-9 HEXAGON and Planet SkySat imagery, our model identifies upcast mounds of underground irrigation aqueducts (*kariz*) with a precision of 87.7%. Across four time slices between 1974 and 2019, the detector identified over 160,000 features within *ca.* 500 km². Beyond familiar patterns of mass destruction and urban or agricultural encroachment from the 1980s onward, diachronic analyses of bounding-box shape, size, and distribution reveal an increasingly fragmented landscape, with clustered features becoming more actively maintained in recent years. We argue that this transformation reflects path dependence and inter-community collaboration in Turpanian oases. Our analyses illuminate how communities negotiate and transform technology amid regime shifts and alternative technological choices. Methodologically, our study explores pathways through which deep-learning models can be productively integrated into archaeological hermeneutics.

Keywords Object detection · KH-9 HEXAGON · Indigenous irrigation · Technological maintenance · Fragility · Central Asia · China

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Introduction

Despite recent development in computer vision and its applications to large-scale mapping of vehicles and buildings in aerial imagery (X. Chen *et al.*, 2014; Xia *et al.*, 2018; Zhang *et al.*, 2020), debates persist within archaeology as to whether automated detection can generate meaningful anthropological and historical insights. On the one hand, past years (2019–present) have seen a proliferation of deep-learning approaches seeking to identify archaeological features such as burials (*tumulus* or *kurgan*), desert fortification (*qasr*), underground irrigation (e.g., *qanat*), and agricultural field systems (Verschoof-van der Vaart & Lambers, 2019; Soroush *et al.*, 2020; F. Chen *et al.*, 2021; Altaweel *et al.*, 2022; Li *et al.*, 2022; Bachagha *et al.*, 2023; Casini *et al.*, 2023; Buławka *et al.*, 2024). These exercises differ from prior machine-learning classifications of multispectral satellite imagery or Airborne Laser Scanning (ALS) (cf. Cowley, 2012; De Laet *et al.*, 2007; Menze & Ur, 2012; Trier & Pilø, 2012), insofar as they adapt *deep* convolutional neural network (CNN) architectures such as Mask R-CNN, U-Net, and YOLO to LiDAR derivatives and optical imagery, often achieving greater accuracy. Nonetheless, while such methods frequently deliver impressive benchmarks in testing and validation, reports remain confined to the sole placement of points or polygons on maps that results in limited engagement with broader archaeological questions (cf. Cowley, 2012). Elsewhere, scholars have—with varying degrees of success—applied deep-learning methods to identify modern threats to archaeological sites using time-lapsed aerial imagery (e.g., Altaweel *et al.*, 2024; Casillo *et al.*, 2025). While this work complements cultural heritage management and preservation as a contemporary industry, its contributions to archaeological interpretations of past socio-material conditions remain unclear.

Indeed, archaeological skepticism towards object-detection models prevails over their accuracy, analytical power, and at times efficiency (e.g., Casana, 2014, 2020; Parcak, 2019). While deep learning has partly addressed criticisms of high false-positive rates, or low precision (Soroush *et al.*, 2020: 2), key issues continue to exist where these models solely add to the “data deluge” without substantive contributions to archaeological theory (Bevan, 2015). Meanwhile, such models perpetuate an epistemological pursuit of archaeological “sites” as predefined, bounded, and discrete units—what Casana (2020: S96) terms “easter eggs”—that have long been problematized by landscape archaeologists (Dunnell, 1992; McCoy, 2020). Following Scott’s (1998) conception of large-scale mapping practices as instrument for homogenization and control, rapid detection at scale risks effacing outliers, local heterogeneity, and community agency to which anthropologists have traditionally paid attention. Furthermore, the deep neural networks underlying these models reinforce concerns over black-box opacity and techno-solutionism, raising questions about reproducibility and academic accountability (e.g., Lambers, 2024).

This paper attempts to reconcile this tension by tracing the diachronic changes in detected features of upcast mounds associated with underground irrigation aqueducts (*kariz*, pl. *karizlar*) in Turpan, Xinjiang, using high-resolution KH-9 HEXAGON and Planet SkySat archives spanning from the 1970s to the present. We

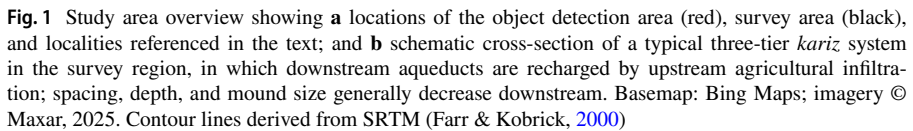
demonstrate that our fieldwork in Turpan (April–October 2023) and case studies of similar technologies elsewhere informed our analysis of model-generated datasets and the resultant interpretation of oasis water management. Given that many of the aqueducts remain in use, we treat these large datasets as an archaeological record—albeit contemporary—of shifting irrigation practices amidst institutional and technological changes that continue to shape the aqueducts today. Through spatial signals, ethnographic observations, and comparative anthropology, we outline in this paper an increasingly fragmented landscape of aqueduct management over recent decades where inter- and intra-mural community collaboration in high-density regions has resulted in the ongoing operation of some *karizlar*.

To our knowledge, our study represents the first attempt to trace such “pixelated transformations” made possible through large-scale object detection in historical very-high-resolution imagery with revisiting capacity. These transformations—“pixelated” insofar as they become visible not through the sole presence of bounding boxes but their subtle and iterative changes over time—reveal contextual management practices that are otherwise inaccessible to archaeologists. Our findings therefore provide a methodological pathway through which automated data can be meaningfully leveraged to address fine-grained anthropological and archaeological questions.

Underground Irrigation Aqueducts in Turpan

Our study focuses on parts of the Chatqal, Yar, Grape Valley, and Ayding Town(ship)s in Turpan that covers a *ca.* 457-km² area where *kariz* aqueducts are most densely distributed (Fig. 1a; XUARBC, 2011). The study area lies at the northern rim of the Tarim Basin, bounded by the Flaming Mountain and *Tuz Tagh* to the north, the *Kum Tagh* to the east, and the *Chöl Tagh* to the south (Stein, 1933). Much of the area sits below sea level, with its lowest point at Ayding Lake, reaching below −150 m (Huntington, 1907). Given this concave topography, the area is extremely arid, receiving an annual precipitation of 16.6 mm against an evaporation rate exceeding 2500 mm per annum (Zhang, 1998). Several perennial streams descending from the northern piedmont and the Tianshan (Tengri) range deposit thick alluvial sediments in the area, within which aquifers are trapped by underlying shale (Barbaix, 2021; Shang, 2020; Shi & Li, 2026). The sharp water-table decline from piedmont to the Ayding Lake contrasts with that of the topography, creating the appropriate conditions for *kariz* operation (Fig. 1b).

Our analysis concerns the Turpanian *kariz* irrigation structures—otherwise known as *qanat*, *falaj*, *foggara*, *karez*, or *khattara*—that often contain an underground gallery accessible through semi-evenly spaced vertical shafts for maintenance (*e.g.*, English, 1968; Wilkinson, 1977; Sala *et al.*, 2010; Yazdi & Khaneiki, 2016; Angelakis *et al.*, 2017; Boucharlat, 2017; Charbonnier & Hopper, 2018). Given the fragile alluvial sediments in Turpan, dredging of these underground galleries takes place annually—if not more frequently—and produces cone-shaped accretionary mounds around the shafts. From above, these upcast mounds



This visibility, alongside the unusual density of *kariz* (over 1600 mounds/km² in parts of Turpan), perhaps explains the *kariz*'s popularity among scientists who attempt deep-learning-based object detection. Traditional machine-learning approaches have struggled to capture the variability in mound morphology across contexts, and Soroush *et al.* (2020) were among the first to apply CNNs to detect upcast features in 1968 CORONA images from northern Iraq. More recently, Buławka and team (2024) employed meter- to sub-meter HEXAGON imagery to detect such features in a range of geographical settings. In Turpan, Li *et al.* (2022) pioneered automated detection using RGB Google Earth mosaics and YOLOv5. Ilniyaz and his team (2025) further developed a topography-informed buffering approach that maps the linear alignments of *karizlar*. Subsequent work of Li *et al.* (2023) combined CORONA-based detection with land use and land cover (LULC) datasets to document significant *kariz* destruction across cropland, bare ground, grassland, and built-up classes. Although valuable for correlating environmental change with *kariz* decline, this latter work largely cast local communities as passive actors in the face of water-table drawdown and

development encroachment. The spatial resolution of CORONA imagery (*ca.* 2 m/px) also precluded observations of fine-grained accretion and weathering at individual *kariz* mounds.

Archaeologically, the Turpanian *karizlar* are generally considered younger than similar aqueducts in Iran, Oman, Egypt, Libya, Syria, and Jordan (Abudanh & Twaissi, 2010; Cremaschi *et al.*, 2018; Goblot, 1979; Lightfoot, 2000; Wilson, 2004). Although some Chinese scholars posit a mainland origin of *kariz*, citing the Han-dynasty Dragon-Head Aqueduct (*ca.* 128–117 BCE), these claims are generally rejected due to functional differences and the absence of a plausible transmission route (cf. Bertrand, 2010; Trombert, 2008). Linguistic affinity, including shared terms *karez* and *kariz* in Persian and Turkic languages, supports a hypothesis of the eastward diffusion of the technology from West Asia. Trombert (2008) proposes an 18th-century introduction of the *kariz*, whereas recent radiocarbon evidence indicates adoption by at least the fifteenth century (Mächtle *et al.*, 2019). Our team's OSL and radiocarbon dating of the aqueduct of Khelip (Fig. 1) predates the latter findings, suggesting the presence of such technology in as early as 11th–13th centuries (Shi, 2025; Shi & Li, 2026).

Many *karizlar* remain in use today, although their numbers are declining: a JICA (2006) report documented an over 40% reduction in functional aqueducts between 1994 and 2006 in the larger Turpanian region, dropping from 586 to 331. In this paper, we examine how shifting management practices and community responses have negotiated technological practices among pronounced environmental and institutional changes from the 1970s to 2019.

Materials and Methods

In April–October 2023, ethnographic and archaeological fieldwork was conducted in the study area, where we surveyed over 60 functional and recently desiccated *kariz* aqueducts (Fig. 1). These ground observations allowed us to characterize the diversity of *kariz* morphologies and maintenance regimes, directly informing image annotations and the interpretations that followed. We performed YOLOv11-based object detection on KH-9 HEXAGON imagery from 1971 to 1984 and Planet SkySat imagery from 2019. Our pipeline is summarized in Fig. 2.

HEXAGON Imagery

KH-9 HEXAGON imagery derives from a series of Cold War reconnaissance missions launched between 1971 and 1986, comprising 19 successful satellites that followed the CORONA (KH-1–4B, 1960–1972) and GAMBIT (KH-7, 1963–1967) programs (Hammer *et al.*, 2022). Although the Panoramic Camera System (PCS) films were declassified in the early 2010s, they were only transferred to the USGS Earth Resources Observation and Science (EROS) Center for ingestion into Earth-Explorer in 2020, and their archaeological applications have only started to proliferate (Ur & Reade, 2015; Ur *et al.*, 2021). Whereas later CORONA missions (KH-4A

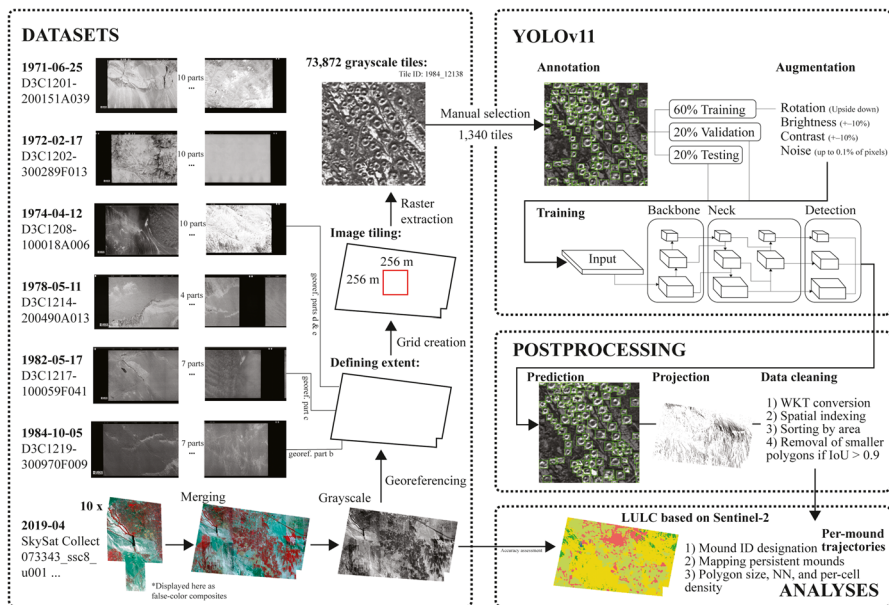


Fig. 2 Object-detection workflow for archaeological features using multi-source, multi-date, very high-resolution satellite imagery. The pipeline includes image acquisition and preprocessing; tiling; YOLO-based inference (architecture simplified from Sapkota *et al.*, 2025); post-processing; and GIS integration

and 4B) achieved post-orthorectification ground sampling distances (GSD) below 2 m in places, HEXAGON PCS products offer substantially finer spatial resolution—often 1 m and sometimes sub-meter—with generally lower cloud cover that allow reliable detections of *kariz* upcast mounds. As with CORONA, our study area benefits from repeated coverage (National Reconnaissance Office, 1982), which permits diachronic analysis at a temporal density rarely attempted in archaeological research. HEXAGON’s panoramic scanning geometry introduces cross-track distortions similar to CORONA imagery (Casana *et al.*, 2012, 2023), but orthorectification and preprocessing mitigate these effects for inter-date comparison.

We acquired six sets of HEXAGON scenes for Turpan via USGS EarthExplorer (Fig. 2). Although winter sun angles cast longer shadows and potentially aid mound identification (Buławka *et al.*, 2024), data availability in our AOI led us to choose spring and autumn scenes with comparable solar geometry. Films were scanned at 7- μ m pixel size and delivered in four to ten grayscale (0–255) parts per scene (cf. Hammer *et al.*, 2022). Due to cloud cover in the 1971–1972 imagery, we georeferenced five image parts (1974, 1978, 1982, and 1984) to UTM zone 45N (EPSG:32645) using a third-order polynomial transformation, with the 1982 scene first processed against Google Earth imagery to serve as the baseline. Each part was rectified using 50–100 ground control points (GCPs), yielding residual RMSEs < 5 pixels (Table 1). We prioritized buildings and *kariz* mound centroids as GCPs, given that roads and geomorphic features in this rapidly developing desert landscape have changed dramatically in the past decades. Mean positional errors were less than 5

Table 1 Georeferenced HEXAGON images

Date	USGS ID	Part	#GCP	Mean error	Resolution	Notes
1974-04-12	D3C1208-100018A006	d	46/51	2.73027	1.029	Upside down; only western half used
		e	56/56	4.2986	1.0052	Upside down
1978-05-11	D3C1214-200490A013	c	46/51	4.053	1.72	Rejected due to poor quality
1982-05-17	D3C1217-100059F041	c	93/101	4.81788	1.0971	Baseline for georeferencing
1984-10-05	D3C1219-300970F009	b	52/53	2.700152	1.2498	High local contrast

pixels and in many cases 3, which—given the *ca.* 1 m or finer GSD—ensure that all upcast mounds can be matched across dates. All georeferenced parts were saved with NoData = -9999 and LZW lossless compression to avoid downstream mosaicking issues; parts d and e from 1974 were subsequently mosaicked in GDAL. Because of the comparatively low resolution of the 1978 scene that precludes direct comparison with other dates (Table 1), we reserved it for reference only.

All rectified scenes were clipped to the spatial extent of available SkySat coverage (see below) and preprocessed in ENVI using histogram equalization with contrast and brightness adjustments to standardize variations prior to detection.

SkySat Collect Archives

SkySat, operated by Planet, is a constellation of 15 satellites that has provided frequent Earth-surface coverage at very-high spatial resolution (0.5 m) since 2016 (Planet Labs, 2025). SkySat Collect products comprise four-band, orthorectified segments (6×5.9 km) assembled from multiple scenes (1080×2560 px) acquired along an imaging strip. Although tasking aims for daily acquisitions, actual capture frequency varies by geography and demand (Roy *et al.*, 2021). Archaeological applications of SkySat remain limited, but scholars have begun to leverage its temporal density to monitor monument preservation (*e.g.*, McGrath *et al.*, 2023). In Turpan, relatively complete, cloud-free coverage for our study area is only available from 2019.

We defined the smallest intersectional extent across the four HEXAGON scenes, processed and selected ten SkySat Collect segments covering this footprint with only the southeast corner missing (Fig. 1). All tiles were acquired in April 2019 (detailed IDs in Supplementary Material 1) and delivered as pansharpened assets with lower resolution near-infrared band matched with the resolution of the panchromatic bands. Images were mosaicked in ENVI with histogram matching enabled and georeferenced based on the 1982 baseline. 52 GCPs were identified, and the mosaic underwent third-order polynomial transformation and resulted in an RMSE of 6.02 pixels. The final mosaic has a GSD of 0.68 m/px. We then converted the mosaic to a single-band grayscale raster using the standard luma resampling to optimize perceived brightness of upcast mounds. A global histogram equalization was applied to enhance low-contrast features.

YOLO

We use a YOLO (“You Only Look Once”) single-stage convolutional detector, which predicts bounding boxes and class probabilities in one single forward pass through the neural network (Fig. 2) and is particularly suited to large-area inference when paired with image tiling (e.g., Redmon *et al.*, 2016; Bochkovskiy *et al.*, 2020). In remote sensing contexts, YOLO models have demonstrated efficiency and strong performance in detecting small features such as pedestrians, wind turbines, and invasive plants (e.g., Chen & Yue, 2018; Pham *et al.*, 2020; Gautam *et al.*, 2025). Because convolutional receptive fields read surrounding pixels within each tile, the detector draws on contextual cues (e.g., *kariz* alignment and sedimentary texture) that help differentiate *kariz* shafts from visually similar features across varied land uses. The recent iteration of YOLOv11 introduces C3K2 feature-extraction blocks in the backbone (Fig. 2), which improve small-object detection relative to earlier versions and, in our case, enhance detection of *kariz* mounds (Sapkota *et al.*, 2025).

All satellite images were partitioned with a fishnet grid of 256×256 m tiles and 10% overlap using QGIS (3.28.8). For each date, this produced 18,468 grayscale tiles. We manually selected 340 tiles per date (with same tile IDs across dates) that are representative of the wide variation in *kariz* size, morphology, and density across built area, bare ground, and cultivated lands. Annotations used a single class (“shaft”) via the Roboflow (1.2.12) interface (roboflow.com); label statistics are summarized in Table 2. Although Buławka *et al.* (2024) mapped two classes (single shafts and shaft pairs) and reported better performance for pairs, our preliminary tests on a subset of the 1982 data using YOLOv11s indicated the opposite in Turpan, with mAP@50 (mean average precision at an IoU threshold of 0.50) of 83.4% for single shafts *versus* 80% for shaft pairs. This is likely due to the great variability in the geometry of Turpanian *kariz* within each aqueduct. Field observations confirmed at least three recurrent mound morphologies that we classify as “shafts”: (i) semi-exposed near-village segments, (ii) discrete mound features, and (iii) large upstream mounds.

Across 1,360 annotated tiles, we labeled a total of 18,532 mounds. Annotation counts (Table 2) vary across the 1974, 1982, and 1984 HEXAGON scenes, primarily due to local tonal contrasts despite similar spatial resolution and the application of global histogram equalization. Although some *kariz* mounds were physically destroyed in 1974–1982, our manual annotation identified fewer features in the 1982 scene because of the relatively higher sun angle and a drier surface that suppressed

Table 2 Summary of annotation

Date	Mission ID	# images labelled	# features labelled
1974-04-12	D3C1208	340	5,040
1982-05-17	D3C1217	340	4,769
1984-10-05	D3C1219	340	5,723
2019-04	SkySat	340	3,000
Total		1340	18,532

tonal cues. In contrast, the 1984 scene preserved more low-contrast edges that allowed our identification of additional mounds. The lower number of mounds annotated in 2019—despite higher spatial resolution of the SkySat mosaic—may indicate substantial destruction of *kariz* mounds between 1984 and 2019. In our interpretations of the results (Sects. "Results"—"Discussion: From Path Dependence to Community Collaboration" below), we consider how aforementioned image quality affects detectability.

Given the modest sample size by deep-learning standards, we split the 1360 annotated tiles into 60%/20%/20% train/validation/test sets. To prevent data leakage (*i.e.*, the same geographical locations appearing in more than one split), we manually assigned each location to a single set. All tiles were resized to 512×512 px for training purposes. We augmented the training set via rotations, brightness and contrast adjustments ($\pm 10\%$), and noise addition (particularly to address film artifacts in HEXAGON imagery), doubling the number of training images. Because most *kariz* features in the study area trend NW–SW, we restricted rotations to 180° (*i.e.*, upside-down flips) in order to preserve orientation context. We trained a YOLOv11-x model initialized with MS COCO weights (Lin *et al.*, 2015) using the Adam optimizer ($\text{lr}_0=0.001$) for a maximum of 300 epochs, stopping after 90 epochs when validation metrics stabilized and no longer improved (Fig. 3).

After training, we deployed the model across the full set of tiles for all four dates ($18,468 \times 4$ tiles). A demonstration of the model and the annotated dataset are provided via a GitHub repository (Ultralytics v8.4.7; Supplementary Material 2). Predictions were mapped back into Mercator coordinates using tile IDs and their attribute tables. To address duplicate detections resulting from tile overlap (*cf.* Buławka *et al.*, 2024), we implemented a Python routine that retains the larger polygon for each overlapping cluster where IoU is above 0.9 (*cf.* Hosang *et al.*, 2017). This was followed by a manual review to remove residual duplicates and obvious artifacts.

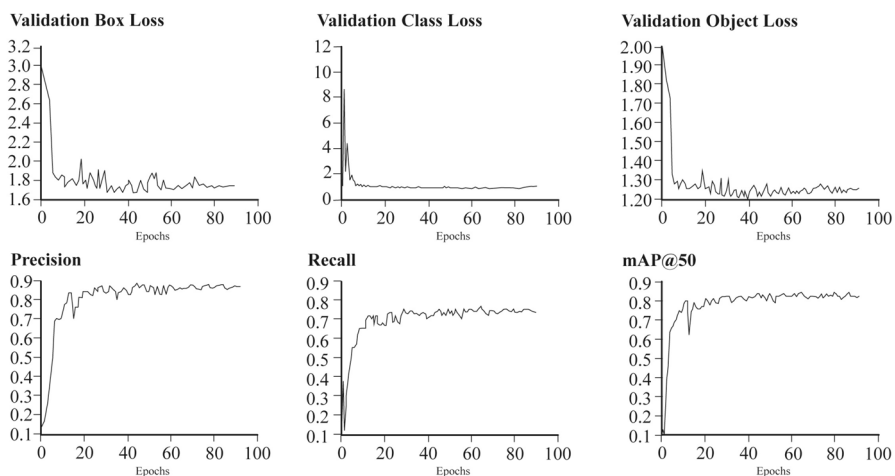


Fig. 3 Model training curves showing convergence of key metrics by the final epoch (90)

LULC Classification

For contemporary LULC classification, we employed Sentinel-2 imagery (45N tile, 10-m resolution) from 2021, following the deep-learning approach proposed by Karra *et al.* (2021), and clipped the dataset to our study area. Modified from similar work in Turpan (Li *et al.*, 2023), we defined four classes: rangeland, built area, crops, and water, where prior classes of rangeland and bare ground were merged. This is mostly due to similar spectral signals of these two classes in our study region and a somewhat arbitrary distinction of “isolated plants” in the rangeland class. In Turpan, the distinction between these two classes bears little significance for *kariz* development (cf. Li *et al.*, 2023). The final raster is presented in Fig. 4.

To evaluate classification accuracy, we used the higher-resolution SkySat mosaic from 2019 for cross-referencing. A set of 500 randomly sampled points were manually labeled based on the SkySat imagery. The resulting classification achieved an overall accuracy of 0.85 (Table 3), with most misclassifications taking place in small LULC pockets like bare ground between cultivated lands (classified as cultivated land in sentinel-based images). In a way, the relatively coarse resolution of Sentinel-2 based classification allowed a more categorical assessment of the general pattern of *kariz* placement than SkySat images. Considering the high accuracy,

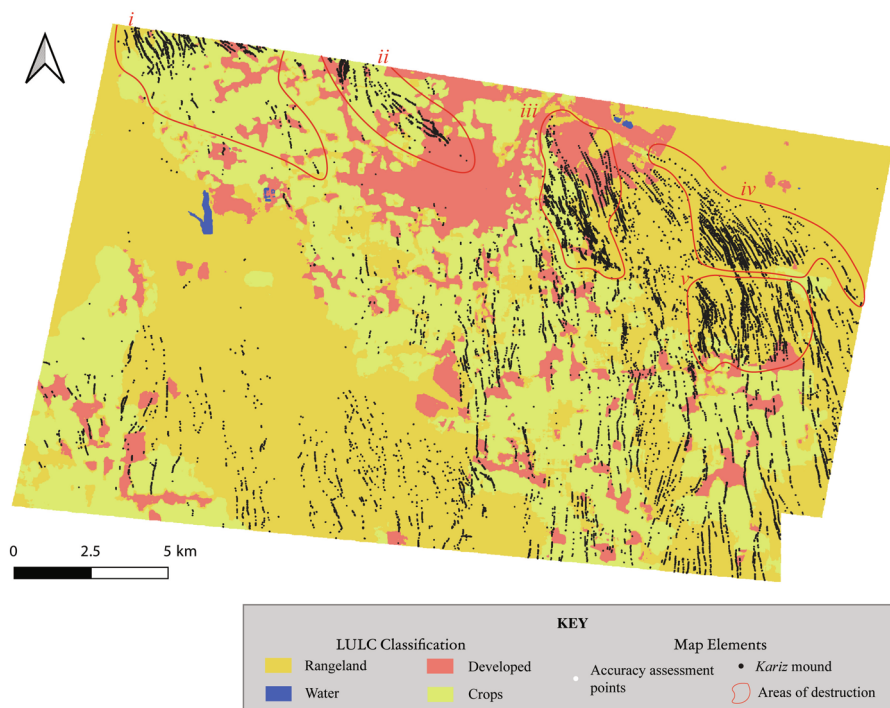


Fig. 4 Sentinel-2-based LULC classification following Karra *et al.* (2021), overlaid with disappeared *kariz* mounds in 2019 and five key areas of mass destruction (i–v)

Table 3 Accuracy assessment of Sentinel-2-based LULC classification (after Karra *et al.*, 2021)

		Classification			
		Water	Crops	Developed	Rangeland
Ground truth	Water	2	0	0	2
	Crops	0	162	9	23
	Developed	0	5	55	11
	Rangeland	0	18	7	206

Sentinel-2 based LULC classification is appropriate for assessing the spatial distribution of *kariz* mounds today.

Results

Model Performance

Our trained object detection model (weights supplied in the repository in Supplementary Materials 2) achieved an average of 87.7% precision, 74.6% recall, and 84.0% mAP@50 across the test set. Model performance is especially strong for relatively large bounding boxes (1024–9216 px²), where mAP@50 reaches 89.3%. Overall, our model is comparable to—or better than—prior deep-learning models applied to similar aqueducts: Soroush *et al.* (2020) reported a precision of 0.654 and a recall of 0.764 for high-density patches using CORONA imagery. Buławka *et al.* (2024) reported a precision of 0.881 and a recall of 0.627 with HEXAGON imagery and YOLOv9. In Turpan, Li *et al.* (2022) obtained a 0.83 precision using YOLOv5 on high-resolution RGB mosaics. Although Li *et al.* (2022) did not report recall, the model in Buławka *et al.* (2024)—which was trained on images from a variety of landscapes and sources just like ours—showed a similar pattern in which recall trails precision. Our detector also slightly outperforms the MMDetection model developed by Ilniyaz *et al.* (2025) who used contemporary Bing imagery (mAP@50=0.837). That our model improvement is incremental reflects the nature of our training data, which results from mixed satellite sources (multiple HEXAGON missions and SkySat). Our model's performance is better for HEXAGON scenes (1972–1984) than the 2019 SkySat Collect mosaic, with precision and recall reported at 0.861 and 0.832, 0.778, and 0.611, respectively. This discrepancy is possibly a result of scene-specific imaging conditions in the SkySat mosaic, especially considering uneven local contrast and variable sun angles in the ten scenes that we ordered (Supplementary Material 1).

For our model, the relative prevalence of false negatives (*i.e.*, lower recall) indicates conservative classification, possibly because of the aforementioned diversity in *kariz* morphology and input images. Manual review of test tiles with low F-1 score (Fig. 5) shows that unusual *kariz* structures—such as those with very closely

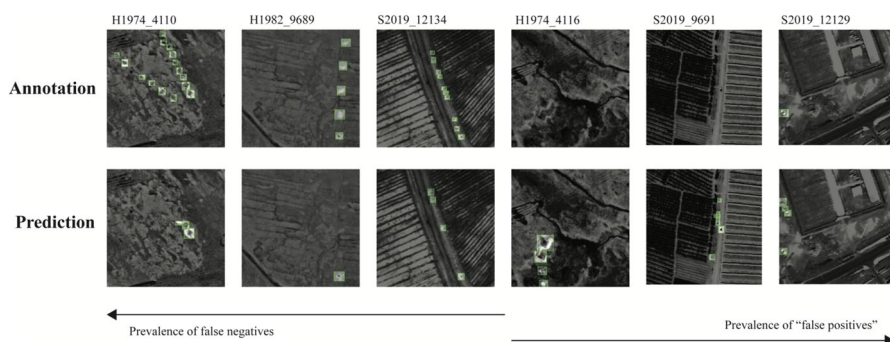


Fig. 5 Test-set tiles with low F1-score (tile IDs above each panel). Top row: human annotations or “ground truth.” Bottom row: model predictions. Low F1-scores reflect false negatives (left panels) and/or false positives (right panels)

spaced shafts or semi-exposed ceilings in near-village segments—are only partially recognized. False positives are rarer: occasionally, linear and regularly spaced circular landforms, especially in the rangeland in the western part of the study area, were misclassified as *kariz* mounds. The model performed well in areas containing other anthropogenic linear–circular patterns (e.g., the edges of greenhouse rows), which we explicitly included as negative training examples. In some places the detector was able to identify features omitted by human annotators (Fig. 5). Given this conservative classifier, we selected modest confidence thresholds (below which the predictions will be discarded) to maximize detection. As summarized in Table 4, a threshold of 0.22 yielded the highest F1-score (0.748) for all four dates. At this threshold, mAP on the test set is 0.818. IoU threshold was set at 0.3.

General Patterns

Model predictions across all four dates are reported in Fig. 6. The detector identified 47,854 mounds in 1974; 43,334 in 1982; 52,259 in 1984; and 25,503 in 2019. These numbers generally track the annotation counts (Table 2), suggesting that the model correctly identifies *kariz* mounds discernible to a human specialist. The apparent decline between 1974 and 1982 is a likely result of the comparative quality of the D3C1217 image—involving hazy contrast and a high solar angle—rather than a broad-scale reorganization of the aqueducts. Meanwhile, the increase from 1974 to 1984 corresponds with observed construction of new *karizlar* during this period

Table 4 Model performance at different confidence thresholds (*conf*)

	F1	Precision	Recall
<i>conf</i> =0.12	0.734	0.729	0.740
<i>conf</i> =0.22	0.748	0.818	0.690
<i>conf</i> =0.32	0.738	0.869	0.641

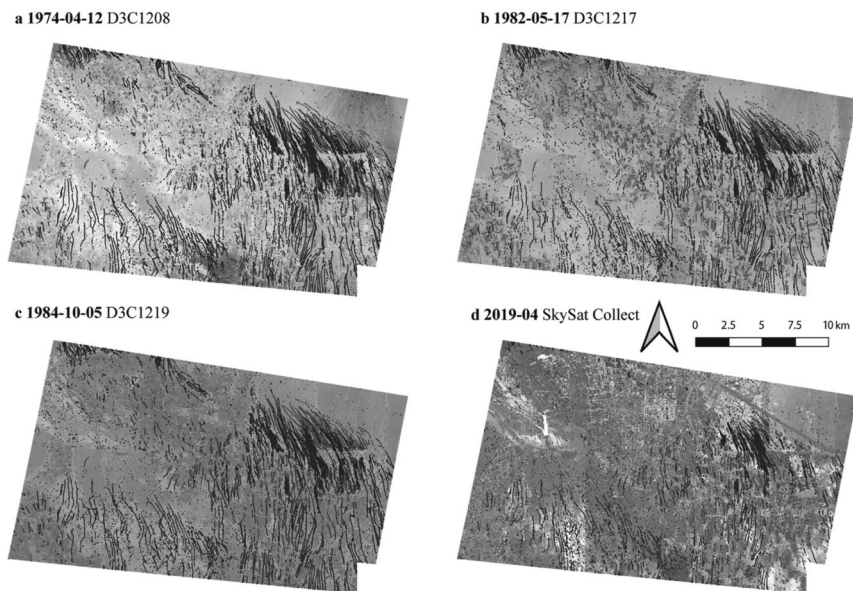


Fig. 6 Model prediction of *kariz* mounds over time, overlaid on satellite imagery from the corresponding years of **a** 1974; **b** 1982; **c** 1984; **d** 2019

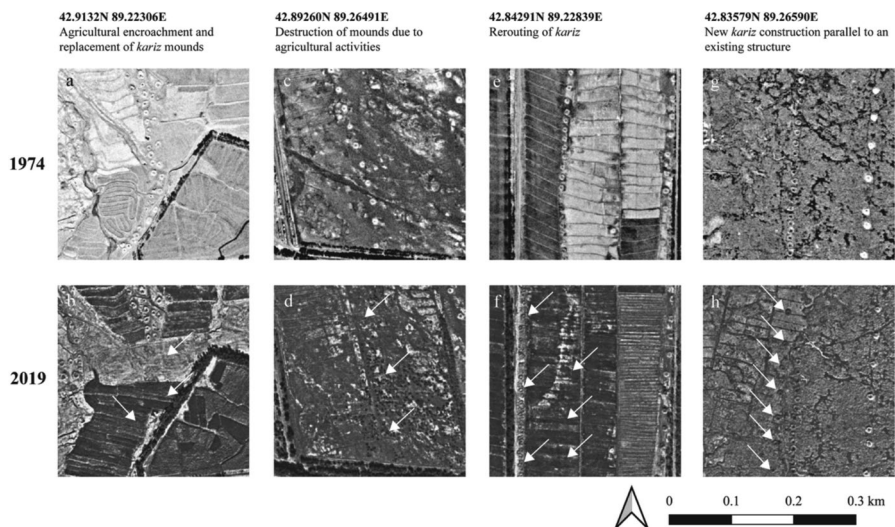


Fig. 7 Destruction (**a–d**) and new construction (**e–h**) between 1974 and 1984; white arrows indicate notable features

(Fig. 7), even as some features were lost. In other terms, the fluctuation of the 1970s and 1980s reflects a relatively stable landscape in which modest expansion and attrition of the irrigation structures took place. The over 50% reduction in mound numbers between 1984 and 2019, however, cannot be explained by image qualities alone—especially given the higher spatial resolution of the SkySat mosaics.

Indeed, overlaying predictions with the LULC classification confirms the loci of mound loss in built-up and cultivated areas in 2019. Based on prediction results, we assigned a unique ID to each of the detected feature. For those that persisted across multiple dates, the IDs were matched by overlap and centroid proximity, yielding a conservative estimate of 27,322 mounds existing across 1974 and 1984. Of those, only 16,211—or 59.3%—survived into 2019. Granted, prediction and georeferencing biases may contribute to undercounting, but the magnitude of mound loss indicates large-scale destruction especially considering the fluctuating numbers in 1974–1984. Among the mounds that persisted through 1974–1984, over 91.2% falling within the contemporary built-up LULC class (1864 out of 2044 detected features) have disappeared by 2019. 72.4% (4799 out of 6626) have disappeared in cultivated land. These LULC classes have often replaced clusters of *kariz* mounds entirely. As shown in Fig. 4, such destruction concentrates in (i) the western reach of Turpan city (NW corner of the study area) with recent agricultural and urban expansion; (ii) Turpan city proper; (iii) eastern Turpan, where a new urban district and administrative offices were relocated near 42.953°N, 89.237°E; and (iv) the G30 highway corridor, where clusters of mother wells used to exist. Notably, some clusters in northern Chatqal County (area v) are absent from the 2019 detections. While part of this absence was due to removal by cultivation, many features show considerable weathering consistent with disuse and the cessation of maintenance that is a result of upstream hydrocarbon extraction, leading to suppressed detection.

In the meantime, only 50.6% of those mounds in rangeland (8953 out of 17,700 detected features) have disappeared by 2019—substantially lower than the overall rate. These findings, in which preservation fares worst in built-up and cultivated areas, align with the observation of Li *et al.* (2023) that introduction of cultivated and urban areas most frequently replace *kariz* aqueducts. It is evident that, in areas where clusters of *kariz* mounds disappear, LULC changes have substantive contributions.

Fragmented Management

Nonetheless, in LULC classes such as rangeland where *kariz* aqueducts tend to persist at higher proportions—at least in their material forms—LULC alone cannot adequately account for their recent abandonment. Further, the mere presence of a surface mound—or a bounding box—does not necessarily indicate functionality of these aqueducts. In fact, field observations suggest that cessation of maintenance is more often registered by mound shrinkage than by the entire removal of mounds. In Fig. 8, we plot changes in mound density, bounding-box area (as a proxy for mound size), nearest-neighbor (NN) distance among mound centroids, and other metrics across dates. NN distance offers a reasonable proxy for shaft spacing within an

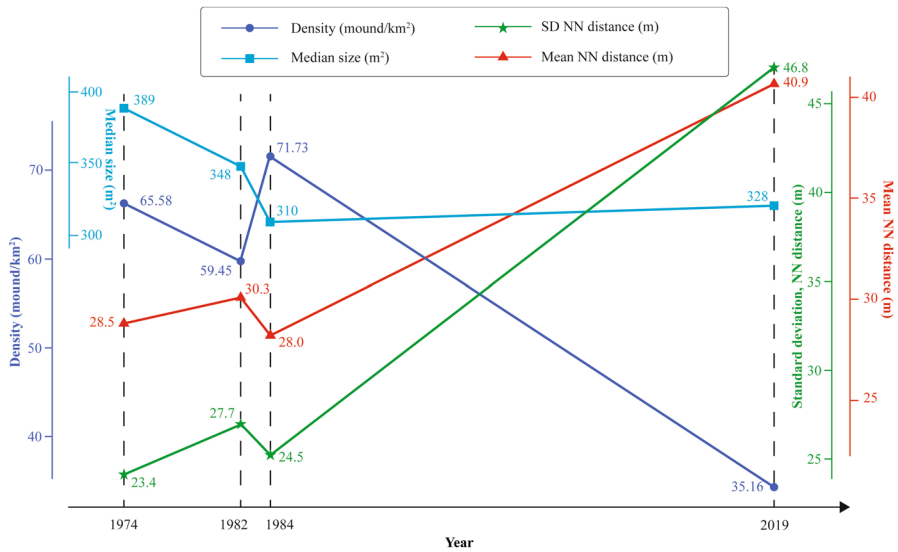


Fig. 8 Diachronic change in *kariz* mound metrics (1974–2019) including mound density, median mound size, mean and SD nearest-neighbor distance

aqueduct because adjacent shafts are typically one another's closest neighbor. This may differ where deeper, widely spaced shafts overlie a shallower, tightly spaced, and usually earlier sequence of mounds, but even in this case the surface features usually belong to the same “system” and overseeing individuals or community. Overall, mound size declines slightly over the years, with the median bounding-box area (with 1% tails trimmed for each year) dropping from 389 m² in 1974 to 348 m² in 1982 and 310 m² in 1984. More strikingly, mean NN distance remained comparatively stable through 1974–1984 (28.5 m, 30.3 m, and 28.0 m) but increases to 40.9 m by 2019. Extant mounds have therefore become both smaller and farther apart.

Nonetheless, if we consider each date's respective distributions of mound size and NN distance (Fig. 9), we see a more complex trajectory of transformations in the irrigated landscape. The standard deviation of bounding-box area remained broadly similar, fluctuating from 102.9 m² in 1982 and 91.3 m² in 1984 to 100.6 m² in 2019. Despite relatively stable mound size over the years, these numbers suggest its generally consistent distribution. NN distance, however, follows a much more volatile trajectory. The standard deviation of NN distance, for example, rose from a relatively stable 14.6 m in 1974, 15.8 m in 1982, and 15.0 m in 1984 to 26.4 m in 2019, indicating greater dispersion—or a “flatter” distribution as shown in Fig. 9—among mound spacings. Likewise, the coefficient of variation (CV) for NN distance increased from 0.82, 0.91, and 0.88 in 1974–1984 to 1.14 in 2019. In other terms, while mound distribution has generally thinned out by 2019, *kariz* became more tightly spaced in some areas and sparse in others, showing greater “heterogeneity” or clustered variability. Given the consistent standard deviation for mound size and the dramatic increase for NN distance, it is unlikely that these shifts in the

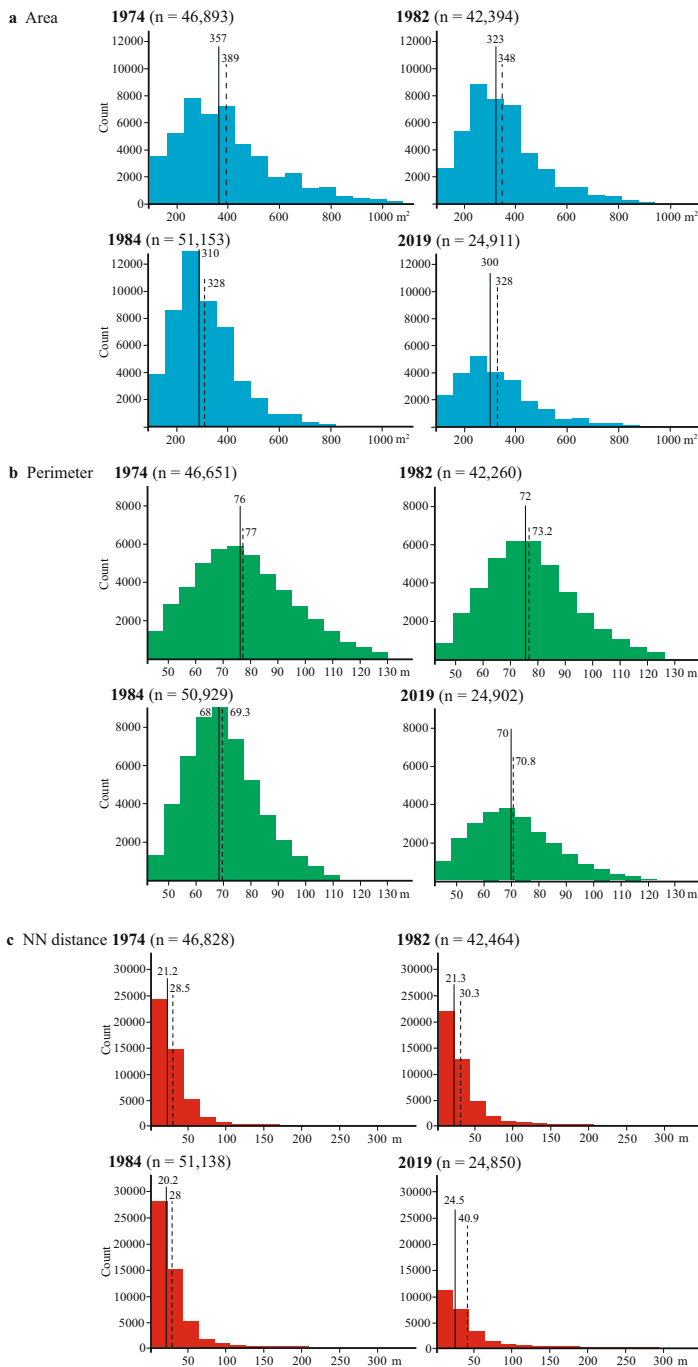


Fig. 9 Per-date distributions of detected polygon **a** size, **b** perimeter, and **c** nearest-neighbor distance, with the outer 1% tails trimmed. Dashed line = mean; solid line = median (50th percentile)

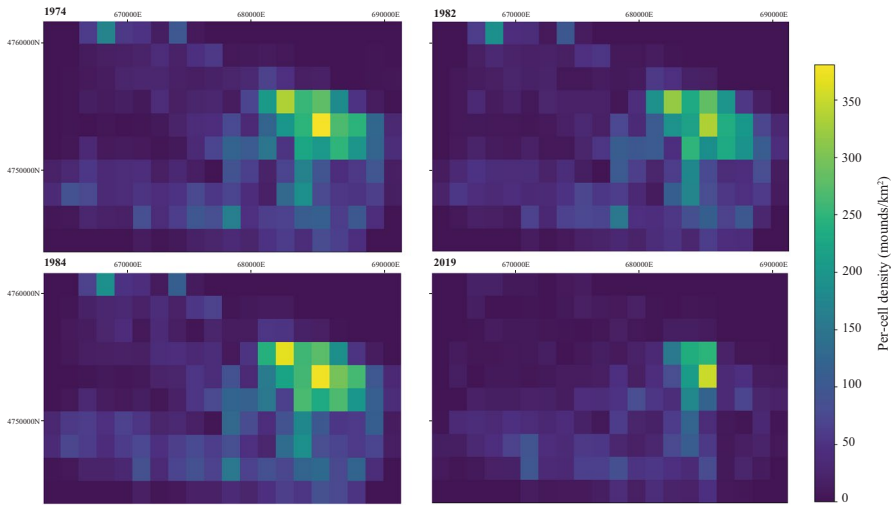


Fig. 10 Changes in cell-wise mound density across the four years in question (1974, 1982, 1984, 2019)

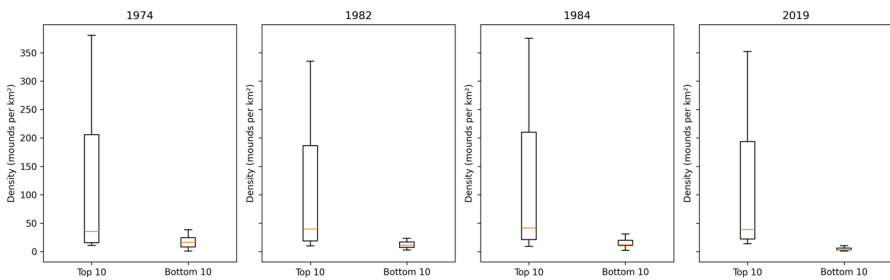


Fig. 11 Mound density in the 10 highest-survivability *versus* 10 lowest-survivability cells

NN distribution are artifacts of our detector. Instead, they reflect a reorganization of *kariz* aqueducts toward an increasingly fragmented landscape in which a few hot-spots where mound density is high have become more prevalent.

As such, we further divided the study area into 200 cells (10 rows $\times$ 20 columns) to evaluate how differently populated cells demonstrate distinct trajectories over the past five decades. Density maps of the four dates (Fig. 10) show that, beyond areas where urban development and agriculture have destroyed clusters of mounds, high-density cells often remained so. We selected the top 10 and bottom 10 cells by the rate of surviving mounds in 2019 relative to 1984, and the top group shows consistently and substantially higher densities than the bottom group (Fig. 11). While mound clusters that were replaced by urban and agricultural land use (Fig. 4) often demonstrate a trajectory from high density to low, mean density in the top 10 cells fluctuates in 1974–1984 and remains high in 2019. Meanwhile, mean density in the bottom 10 dips from 1974–1984 to 2019, once again indicating a more differentiated “moundscape” in which hotspots of *kariz* remain “hot” and formerly sparse

areas become more so. These results mirror the NN-distance patterns as well as the increasingly positively skewed size distribution in 2019 (Fig. 9), where a majority of small mounds became smaller and a few larger mounds “held out.” Given that active maintenance increases mound size, and desiccated mounds gradually shrink, our findings suggest that a few mounds remain actively maintained while many have been abandoned. Taken together, the evidence points to increasingly fragmented management strategies in Turpan by 2019: some “hotspot” communities continue to maintain their aqueducts or even more so, whereas communities elsewhere appear to have given up on their underground aqueducts altogether.

Discussion: From Path Dependence to Community Collaboration

Changes in mound size and spacing evident in the multi-date high-resolution imagery point to shifting regimes of oasis irrigation. Maintenance activities appear to have fluctuated—and in places intensified—between 1974 and 1984. New construction demonstrates a typical “discrete-mound” morphology, with medium sizes (*ca.* 10–20 m diameter) and gaps (up to 30 m), contributing to the slightly decreased size deviation in 1982 and 1984 (Figs. 7 and 9). This pattern generally aligns with archival materials (XUARBC, 2011) documenting *kariz* addition during this period, as well as Rishit’s (2016) observation that agricultural activities remained relatively steady in Turpanian oases during the People’s Commune period between land reform (1950s) and the adoption of the Household Responsibility System (early 1980s). The latter event introduced neoliberal incentives that shifted agricultural priorities towards high-profit cultivars such as grapes, melons, and legumes whose irrigation needs might not be readily met by *kariz* alone (*cf.* Barbaix, 2021).

Indeed, the years between 1984 and 2019 saw a general decline in *kariz* maintenance and use, potentially replaced with pump wells, resulting in greater proportion of smaller *kariz* mounds. Urban and agricultural development replaced some clusters or “hotspots” entirely—especially around Turpan City (Fig. 4)—aligning with the observations of Barbaix (2021) and Li *et al.* (2023). Maintenance activities remain or intensify in hotspots that persist, whereas communities with fewer aqueducts seem to have abandoned *kariz* entirely and instead relied upon wells or state-sponsored surface water infrastructure (*e.g.*, 17 reservoirs in the Turpan Prefecture as of 2013; The World Bank, 2013).

Scholars working on similar shaft-and-gallery underground aqueducts in other regions have documented their recent disappearances with close attention paid to the causes and effects of such desiccation (*e.g.*, Lightfoot, 1996a, 1996b, 1997; Wessels & Hoogveen, 2002; Nasiri & Mafakheri, 2015; Chiotis, 2017; Khan *et al.*, 2017; Manuel *et al.*, 2018; Bensi, 2020). Many have attributed causality to water-table drawdown as a result of pump wells and anthropogenically induced droughts. This is often supplemented with a discussion on the ideological effects of “modernization,” whereby younger generations are less inclined to maintain aqueducts given reliable pump well water (*cf.* Barbaix, 2021). Elsewhere, researchers like Cressey (1958) emphasized episodic disruption by extreme weather events, such as flash floods. On the other hand, historical analyses show how changing institutional priorities

could shape the operation of the irrigation infrastructure. In Oman, again, Wilkinson (1977) argues that tribal codes variably impeded the spread of *aflaj* relative to pre-Islamic eras, while Cremaschi *et al.* (2018) describe how illegal seizure and bans on produce from the Salut oasis precipitated *falaj* desiccation by the end of the eleventh century CE. In our team's work elsewhere (Shi & Li, 2026), we similarly show how cultural heritage preservation may inadvertently suspend routine *kariz* maintenance. While our findings here discuss an episode of *kariz* desiccation, we show further how such desiccation has an uneven spatial distribution over just five decades, in which local management styles have played a significant role in shaping where the technology persists and deteriorates.

Our analyses of changing distributions of mound density, size, and spacing from the 1970s to 2019 indicate increasingly fragmented management strategies in Turpan, where areas of prior high concentration received more attention, and dispersed aqueducts were largely abandoned. A traditional reading would invoke theories of path dependence: regions of higher *kariz* density may have experienced “lock-in,” where agricultural rhythms, socio-cultural fabric, embodied social memory, and especially material infrastructure are intricately connected with the aqueducts (Arthur, 1989; Hegmon *et al.*, 2016; cf. Gupta, 2018; Hodder, 2024). A corollary is that the introduction of new technologies, such as pump wells, could hardly disrupt the regional reliance upon *karizlar* entirely. Further, the increasing degree of heterogeneity (*i.e.*, growing differences among sites) resonates well with arguments that rising “complexity,” specialization, or dispersion in material culture, can precede technological or systemic collapses (Flannery, 1972; Tainter, 1988). A more recent literature on Early-Warning Signals (EWS) similarly links increased variance and unequal material distribution to declining resilience and regime “reorganizations” (Downey *et al.*, 2016; Scheffer *et al.*, 2021; Spielmann *et al.*, 2016). In a way, this work describes a pattern akin to what we have observed, where increasingly heterogeneous decisions with respect to *kariz* takes place as the traditional technology faces regime and technological shifts.

Nonetheless, this scholarship on complexity and resilience describes technological change during crisis but rarely *explains* it beyond a historiographical model. Further, proposing a directionality of socio-material trajectories risks effacing local agency in shaping technological change. Our ethnographic fieldwork in areas of high *kariz* density (Fig. 1a) suggests that hotspots often show extensive degrees of intra- and inter-mural community collaboration compared to other areas (cf. Shi, 2025: 118–164). For individual communities within these hotspots, irrigation quotas are distributed relatively equitably. While Al-Marshudi (2007) reported that *aflaj* in northern Oman typically lease water by time—and similar practices are documented for most regions in Iran (Khaneiki, 2019; Yazdi & Khaneiki, 2016)—Turpan demonstrates a mixture of area- and time-based pricing that varies by oasis. Regardless of the measured units, villages organize hydrological committees that record water use and assess fees dedicated to *kariz* maintenance (Fig. 12a, b). Because underground galleries are easily blocked by the unstable alluvial sediments and require annual cleaning at considerable cost, it is not in the interest of head-enders (upstream users near the *kariz* reservoir) to appropriate all available flow, as they lack the labor and resources for future repairs. Farmers within *kariz*-owning communities “bargain”



Fig. 12 A collection of photographs showing: **a** a village bulletin board listing irrigation fees owed (photo taken by the authors); **b** a plaque shown by local farmers indicating funding for *kariz* maintenance (photo taken by the authors and cropped to protect identities); and **c** rerouting of *Kum Kariz* to increase its flow rate; source: Google Earth, Maxar Technologies (2008)

over maintenance duties proportional to water usage to maximize collective benefits. Elsewhere, Khaneiki (2019) has observed similar patterns in Meymand, Iran, where high costs for *qanat* construction and maintenance necessitate a form of “territorial cooperation” in which resources are pooled. For Meymand, the construction of *qanat* would also involve an endowment (*waqf*) to public works such as a bathhouse that can be shared among community members.

Beyond collaboration within individual villages, inter-village collaboration allows the “hotspots” that we observe to remain so. At *Kum Kariz*, for example, farmers connected their aqueduct with the neighboring *Khelip Kariz* to address upstream blockage due to prolonged neglect (Fig. 12c). A local expert at the *Khelip Kariz* described to us his *asmi-kara* (“straight-work”) technique in which wooden scaffolding (*jiazi*) was used to hang sticks and survey straight lines between shafts of these two neighboring *karizlar*. Such projects, as he explained, regularly facilitate knowledge sharing across villages. At *Tur Kariz*, a village elder recounted how other communities with drying aqueducts had to send for specialists from “successful” villages such as his own. Likewise, a *kariz* worker at *Qiyishlar Kariz*, having maintained *kariz* since 1992 until his retirement at the age of 63, noted that he would work for other villages for 250 yuan per day instead of his normal rate of 150 yuan. While comparable exchange of knowledge was observed in Iran (Khaneiki, 2009), the underlying principle in Turpan is that these villages would now co-sponsor and sometimes co-manage multiple aqueducts, contributing labor or funds to what one may call collective maintenance. As in Meymand, our survey in Turpan reveals a multi-tier arrangement (Fig. 1b) in which downstream *karizlar* depend on infiltration from upstream agriculture that is supported by further upstream *karizlar*. In such cases, downstream or southern communities sometimes subsidize upstream maintenance. For instance, at *Tur Kariz* the same elder reported selling lands of forestry to downstream communities, with the proceeds reinvested in extending their own aqueduct by one *tashma* (the distance between wells spacing, ca. 30 m in this case) in order to increase the flow rate.

Viewed as such, the endurance of *karizlar* in the “hotspots” that our analyses identify can be read as community-level collaboration necessitated by recurrent

repair needs that upstream users alone cannot meet. While ethnographers in Iran and Oman have noted such local collaborations, the present study is to our knowledge the first to visualize their spatial signatures via satellite images (Figs. 6 and 10). From the 1970s to the early 2000s, anthropologists have paid particular attention to the “indigenous,” acephalous, and diverse irrigation management that contradicts Hardin’s (1968) “tragedy of the commons” conjecture in which a centralized authority is posited to prevent head-enders from monopolizing flows (Geertz, 1972; Hunt *et al.*, 1976; Coward, 1977; Hunt, 1988; Netting, 1993; Mabry, 1996; Trawick, 2001). Less widely known is that Nobel laureate Elinor Ostrom—celebrated for her work on self-governing entities—was perennially interested in how farmer-owned irrigation challenges the “state-of-nature” game theory that one rational party will invariably hog water (Gardner & Ostrom, 1991; Weissing & Ostrom, 1991; Ostrom, 1992; Ostrom *et al.*, 1992). Citing extensive empirical evidence, her work emphasizes autonomy (control over water) and proportionality (equal water rights and maintenance duties) that sustain the productivity and ongoing operation of many local irrigation systems (Ostrom & Gardner, 1993). In these examples, state grants for maintenance may eliminate incentives for head-enders to solicit assistance from downstream water users. Implicit here is the fragile materiality of irrigation infrastructure—in Nepal, Peru, and, in our case, borderland China—that demands substantial and recurring labor and expense, and thereby cross-household and cross-village contributions. For the Turpanian *karizlar* in particularly unstable geomorphological conditions where alluvial and loess sediments can collapse and block the underground galleries, such community-level collaboration—visible in the spatial history that we outline here—is how well-maintained clusters can continue to exist.

Concluding Remarks

Previous applications of deep learning in archaeology have focused mainly on the placement of points or polygons on maps and reporting (usually impressive) benchmarks, often without sufficient deliberation on their implications for the organization, operation, and evolution of human societies. In this paper, we used YOLOv11 alongside very-high-resolution satellite mosaics—three declassified and one contemporary—to trace small-scale changes in *kariz* size, spacing, and density at a scale inaccessible to manual survey alone. With an average precision of 87.7%, our trained detector identified 168,950 features across four time periods from 1974 to 2019. While these features indicate a familiar pattern of destruction and urban/agricultural encroachment seen in similar cases, our fine-grained—or “pixelated”—analyses reveal an increasingly fragmented landscape, where regions that had higher densities tend to maintain aqueducts while more dispersed regions have largely abandoned *kariz*. In addition to a traditional view of “path dependence,” we argue that local collaborative management also plays a role in sustaining these “hotspots.”

Our study shows that for deep-learning applications to be analytically and anthropologically meaningful, fieldwork and comparative anthropology remain important. Extensive ground survey and participant observation in Turpan helped us recognize

kariz mounds across a wide range of forms and shades in satellite images and motivated our attention to size, spacing, and edge sharpness as signatures of maintenance strategies. Fieldwork and case studies elsewhere also provided comparable data points that are necessary to interpret model outputs beyond a mechanical theory of path dependence, highlighting the contingent and negotiated character of local technological choices (cf. Lemonnier, 2002). Finally, by integrating automated detection with social theory, we outline a pathway through which the HEXAGON archive—with its combination of high spatial resolution and repeated coverage—can contribute to documenting and interpreting the comparatively recent destruction and preservation of archaeological landscapes.

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Author Contribution CHS: Conceptualization, Data curation, Funding, Formal analysis, Investigation, Visualization, Project administration, Writing—original draft, review and editing. CEH: Data curation, Investigation, Validation, Writing—review and editing.

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Data Availability Datasets and trained model are provided in Supplementary Material 2.

Declarations

Ethics Approval The part of the study that involves human subjects was reviewed and approved by the institutional review board of Stanford University (eProtocol number: 68421).

Consent to Participate Informed consent was obtained from all contemporary participants.

Conflict of Interest The authors declare no competing interests.

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